MATTER-FIRST AUTOMATED DISCOVERY OF TRANSIENT WARP PRECURSORS

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We present an automated matter-first pipeline for discovering warp-precursor spacetimes: an optimizer proposes matter, a constraint solver (`GRTresna`) builds initial data, a GPU evolution code (`GRTeclyn`) advances the full 3+1 Einstein equations, and null geodesics integrated through the *time-evolving* metric test whether light arrives earlier than in flat space. MAP-Elites quality-diversity search over a 39-dimensional bicomplex Q-ball genome saturates at candidate 87 ($S = 825$); CMA-ES refinement raises the score to 852 (candidate 146) by improving confinement-gated persistence. In the headline validation run ($L = 128$, $N = 256$, three AMR levels, $t = 30$), null rays arrive 20.2% earlier than the matched flat path (5/5 rays reaching, all quality gates passed). A three-run Richardson ladder (coarse/medium/fine, $N = 192/256/384$) yields 20.4%/20.2%/20.2% with fine–medium relative difference $\ll$1%. An external ignition controller switches off at $t = 4$; because the headline geodesic launches at $t_{\text{emit}} = 4$, the 20.2% advantage is measured in freely evolving Einstein–Klein–Gordon dynamics. Matter confinement falls to $\sim$11% by $t = 30$: the shortcut is transient, not a passenger drive.

I. INTRODUCTION

The goal of this work is an *automatic discovery pipeline* for faster-than-light (FTL) warp precursors. We define a warp-precursor as a constraint-satisfying matter configuration that generates a transient, effective null timing advantage prior to its own dispersal: null rays between prescribed asymptotic endpoints traverse the curved spacetime in less coordinate time than the matched flat-space path. Rather than hand-crafting a metric and hoping the implied stress-energy is physical, we want a closed loop that proposes candidates, evolves them under the Einstein equations, and retains only those that pass a gauge-invariant transport test.

The standard route is *geometry-first* (equivalently metric-first): one prescribes $g_{\mu\nu}$, reads off $T_{\mu\nu} = G_{\mu\nu}/8\pi$, and studies frozen slices [3]. That recipe is poorly suited to automated search. The implied matter is rarely the output of a consistent initial-value problem, and under dynamical evolution it typically disperses or collapses, destroying the intended channel. For a quality-diversity optimizer that must evaluate hundreds of candidates, a geometry-first genome is especially hostile: most deformations of $g_{\mu\nu}$ either violate the constraints, have no local matter realization, or fail as soon as time advances—so the archive fills with numerical failures rather than competing mechanisms.

We therefore invert the loop (Fig. 1). The optimizer proposes matter; `GRTresna` (a constraint solver that returns initial data satisfying the Hamiltonian and momentum constraints) builds the initial slice; `GRTeclyn` [8, 9] (a GPU-accelerated, AMReX-based CCZ4 evolution code) advances the spacetime; and probes ask whether a null shortcut appears. In this paper "FTL" and "null shortcut" mean the same operational claim: an *effective* reduction of null travel time relative to flat space between the same asymptotic endpoints—never local superluminal motion outside the light cone. The fractional advantage is f_{geo} [Eq. (10)]; for example $f_{\text{geo}} = 0.20$ means the curved transit is 20% shorter than the flat comparison.

That criterion is load-bearing, not cosmetic. A warp precursor is useful only if it changes what light can do between fixed distant observers. Local coordinate signal speeds, lapse/shift cosmetics, or a positive frozen-slice diagnostic can look "superluminal" while no ray actually connects the endpoints sooner once the metric evolves. Asking whether null geodesics arrive earlier than in flat space is therefore the operational definition of the search target: a gauge-invariant transport certificate that the optimizer must earn, not a post-hoc plot.

Equally important is the *scoring metric* that turns each evolution into the fitness $S(\theta)$ used by the archive [Eq. (6)]. MAP-Elites converges to whatever S rewards. In earlier campaigns with weaker objectives the search reliably *cheated*: it harvested coordinate proxies while matter dispersed ("disperse-then-warp"), banked frozen-slice positives that failed the evolving geodesic test, or climbed scores through numerical pathologies— early horizons, integrator crashes, and other simulation-failure modes that accidentally produced large proxy signals. The present objective therefore rewards only the evolving 4D shortcut and persistence, gates health terms by nontriviality, penalizes exotic excess and horizon formation, and dispersion-gates transport credit so a flying-apart cloud cannot buy a high score. Without that metric design the archive does not discover physics; it discovers loopholes.

The search engine is MAP-Elites [7], a quality-diversity algorithm that keeps one elite per cell of a behavior archive rather than a single global optimum. We prefer this over a pure CMA-ES hill-climb, which collapses onto one basin and discards diverse near-misses, and over reinforcement learning, which would require dense credit

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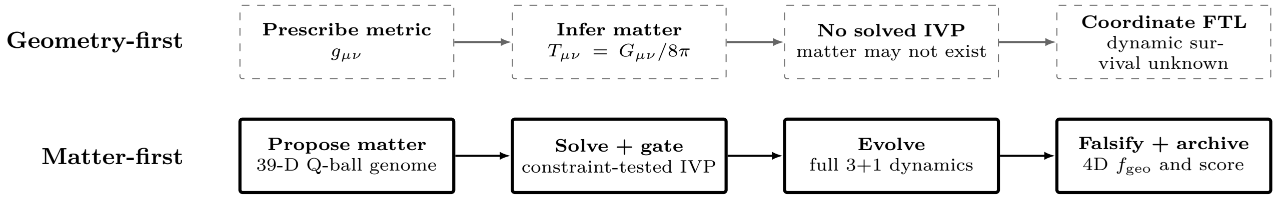

FIG. 1. Geometry-first versus matter-first discovery. Dashed: prescribe a metric and infer its stress-energy. Solid: propose matter, solve and test an initial-value problem (IVP), evolve it, and archive only candidates that pass the 4D null-transport test.

assignment across expensive GPU evolutions with sparse, delayed rewards. After MAP-Elites identifies a high-scoring basin, CMA-ES warm-starts from the archive champion and locally refines the genome before any high-resolution replay or resolution ladder (Sec. III). Both discovery stages run on a deliberately coarse grid chosen for evaluation throughput; production claims are deferred to the later HQ / resolution matrix (Sec. III).

A. Methodological Rigor

Three pillars distinguish this pipeline from geometry-first studies:

1. **Initial value problem:** constraint-clean data from physical scalar sources, not coordinate scripts.
2. **Dynamic feedback:** full BSSN/CCZ4 evolution; dispersing matter collapses the channel.
3. **4D geodesic falsification:** the headline diagnostic is `ftl_geo_evolving`—null rays integrated through the *time-evolving* metric stack. Frozen-slice geodesics are retained as diagnostics but are not the transport certificate.

B. Contributions

(i) An end-to-end matter-first discovery pipeline—MAP-Elites proposal, CMA-ES local refinement of the archive champion, constraint solve, GPU evolution, and matched evolving-geodesic scoring—with an explicit reward rule applied to every evaluation (Secs. II–III).

(ii) A hierarchical campaign protocol: quality-diversity search, CMA-ES hill-climb on the top genome, then longer higher-resolution replay and a three-run Richardson resolution ladder (RC/RM/RF = coarse/medium/fine at fixed domain, varying grid spacing $h = L/N$) only on the refined champion (Sec. V).

(iii) Explicit numerical limits of the surviving candidates: dispersing exotic matter and a transient ignition-phase controller—not passenger transport (Sec. VI).

II. MATTER-FIRST PIPELINE

A. Two-code toolchain

Two open numerical-relativity codes carry the loop. `GRTresna` is a constraint solver: given a matter configuration it returns constraint-satisfying initial data $(\gamma_{ij}, K_{ij}, \phi, \Pi)$ on the initial slice, so that the Hamiltonian and momentum constraints hold before any evolution. `GRTeclyn` [8, 9] is a GPU-accelerated, AMReX-based code that evolves those data forward with the CCZ4 formulation of the Einstein equations under moving-puncture gauge. Neither code is told what metric to produce; the geometry is whatever the proposed matter sources.

B. Search Loop

One evaluation maps a parameter vector θ to a scalar fitness: $\theta \to$ initial data $\to$ `GRTeclyn` $\to$ diagnostics $\to S(\theta)$. The optimizer proposes θ, `GRTresna` solves the constraints, `GRTeclyn` evolves the spacetime on GPU, and post-processing probes reduce the evolution to the fitness $S(\theta)$ that steers the next proposal.

The matter is a canonical complex scalar field ϕ with conjugate momentum Π and potential $V(\phi)$. In the 3+1 split, a scalar field contributes an energy density ρ and a momentum density S_i that act as the sources in the Hamiltonian and momentum constraints,

$$\rho = \tfrac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \tfrac{1}{2}\Pi^2 + V(\phi), \qquad S_i = -\Pi\,\partial_i\phi, \quad (1)$$

where ψ is the conformal factor and $\tilde{\nabla}$ the conformal spatial gradient. The three terms in ρ are, in order, the gradient (spatial shape), kinetic (time variation), and potential (rest) energy of the field; S_i is nonzero only when the field oscillates in time ($\Pi \neq 0$) with a spatial gradient, which is what lets a moving scalar lump carry momentum and drive a shift. Equation (1) is therefore the precise statement of "the matter tells the geometry what to be": θ sets ϕ and Π, and `GRTresna` returns the (γ_{ij}, K_{ij}) consistent with these sources.

The word "canonical" describes the Klein–Gordon equation of motion, not the sign with which every initial lump gravitates. For each lump the optimizer stores

$x_{\mathrm{ex},k} \in [0,1]$, but configuration construction rounds it to $e_k = \mathrm{round}(x_{\mathrm{ex},k})$ and sets $s_k = (-1)^{e_k}$. Thus there is no physical fractional interpolation. In the `GRTresna` constraint solve the signed independent-lump approximation is

$$\rho_{\mathrm{ID}} = \sum_k s_k \rho_k, \qquad (S_i)_{\mathrm{ID}} = -\sum_k s_k \sum_{A=1}^{2} \Pi_{A,k} \partial_i \phi_{A,k}, \tag{2}$$

with inter-lump stress-tensor cross terms omitted. The profiles themselves are then superposed into two painted complex fields—one canonical, one phantom—by the bicomplex model. Candidate 146 rounds to $s_k = (+1, -1, -1, +1, -1)$: two canonical and three phantom sources, with these signs preserved in both the constraint solve and the dynamical evolution.

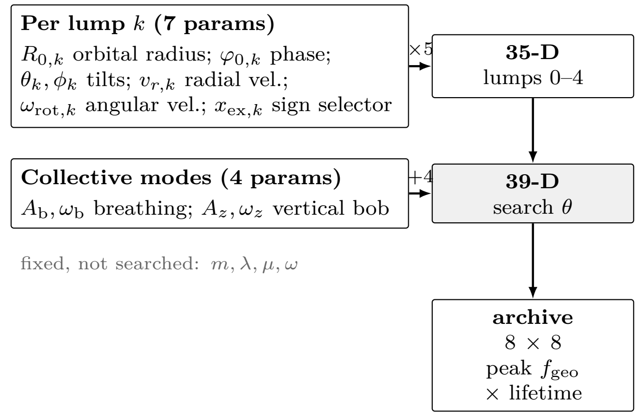

FIG. 2. Search parameterization of the trajectory Q-ball shell: $5 \times 7 = 35$ per-lump placement parameters plus 4 collective modes give the 39-D genome θ. MAP-Elites stores one elite per cell of an 8×8 archive indexed by peak shortcut and FTL lifetime. Q-ball microphysics is held fixed.

C. Q-ball building blocks

The matter is built from Q-balls, and this choice was forced on us by failure. Earlier ansatz families—spherical-harmonic scalar shells and free massive (Klein–Gordon) wave packets—have no bound state and simply spread: their structural persistence collapses to a few percent well before the end of the evolution, taking any transient shortcut with them (Sec. VI A). A Q-ball is a localized, non-topological soliton of a complex scalar field: the field carries a conserved Noether charge under a global phase symmetry, and that charge holds a compact lump of field together against dispersal. This self-binding is precisely what the earlier families lacked. We choose a *complex* rather than real scalar for two reasons. First, only a complex field has the phase symmetry that supports a stable charged lump. Second, its harmonic phase rotation $\phi \propto e^{-i\omega t}$ makes $\Pi = \dot{\phi} \neq 0$, so by Eq. (1) the lump carries a momentum density S_i and can source a non-trivial shift—the ingredient a transport channel needs. Each lump starts from a spherically symmetric equilibrium profile obtained by solving the radial ODE for a sextic potential,

$$V(|\Phi|) = \tfrac{1}{2} m^2 |\Phi|^2 - \tfrac{1}{4} \lambda |\Phi|^4 + \tfrac{1}{6} \mu |\Phi|^6, \tag{3}$$

with fixed parameters $m = 1$, $\lambda = 640$, $\mu = 85333$, $\omega = 0.8$. The attractive quartic ($\lambda > 0$) provides self-binding; the repulsive sextic ($\mu > 0$) is *required* for a stable three-dimensional Q-ball—a pure quartic is the critical case and either collapses or disperses. The ratio λ^2/μ sets the minimum internal frequency $\omega_{\min} \approx 0.316$; $\omega = 0.8$ lies in the thick-wall regime where the equilibrium soliton decays to zero within $\sim$7 code units, making it compact enough to fit multiple lumps inside the evolution box. These microphysical constants are not searched.

D. Shell Parameterization

The search vector has 39 dimensions because the arrangement of the lumps, not the lump microphysics, is what is optimized. The shell contains five lumps—the maximum supported by the solver's state layout—giving enough independent tilted orbits to produce multi-stream frame dragging while keeping the search space tractable. Each lump carries seven searchable placement parameters—orbital radius R_0, phase, two tilt angles, radial velocity, angular (rotation) velocity, and a source-sign selector—giving $5 \times 7 = 35$ per-lump dimensions. Four collective variables set the shell's breathing amplitude and frequency and its vertical oscillation amplitude and frequency, for $35 + 4 = 39$ in total (Fig. 2). Although sampled continuously, the sign variable is applied as the binary selector above; it lets the optimizer choose which initial-data lumps source positive or negative stress-energy.

E. Lump orbit parameterization

Each lump moves in its own tilted orbital plane. The support center of lump k is prescribed by

$$\begin{aligned} r_k(t) &= \max[r_{\min}, R_{0,k} + v_{r,k} t + A_{\mathrm{b}} \sin(\omega_{\mathrm{b}} t)], \\ \varphi_k(t) &= \varphi_{0,k} + \omega_{\mathrm{rot},k} t, \qquad z_{\mathrm{b}}(t) = A_z \sin(\omega_z t), \end{aligned} \tag{4}$$

followed by rotation into the laboratory frame through the two tilt angles (θ_k, ϕ_k). Thus each lump spirals radially (drift $v_{r,k}$), rotates in its own plane (angular velocity $\omega_{\mathrm{rot},k}$; negative values are retrograde), and shares a collective radial breathing $(A_{\mathrm{b}}, \omega_{\mathrm{b}})$ and vertical oscillation (A_z, ω_z). The range of possible orbits includes retrograde and prograde configurations, fast and slow rotation, radially drifting spirals, and nearly static scaffolds.

F. Trajectory controller (PD pump)

During evolution, a proportional–derivative (PD) controller drives the two real components of each complex field toward a moving target profile at the prescribed support center:

$$\begin{aligned}\partial_t \Pi_A|_{\rm PD} = &-w k_p(\phi_A - \phi_A^\star)\\ &- w k_d(\Pi_A - \Pi_A^\star), \qquad A = 1, 2,\end{aligned} \tag{5}$$

with gains $k_p = 12$, $k_d = 7$, and w the product of a localized envelope and a constraint-sensitive governor. This is an external, non-Lagrangian source in the scalar momentum equations; no corresponding pump stress-energy is added to $T_{\mu\nu}$, so $\nabla_\mu T^{\mu\nu} = 0$ is not exact while the pump is active. The pump switches off at $t_{\rm pump} = 4$; after that, the matter evolves freely under Einstein–Klein–Gordon dynamics. The scoring function credits 4D geodesic transport only for emission times $t_{\rm emit} \geq t_{\rm pump}$, so the headline shortcut is always measured in the free evolution window.

G. Constraint gate and source-model consistency

A post-load constraint gate rejects inconsistent initial data before any GPU time is spent: the `GRTresna` Hamiltonian and momentum residuals must fall below 5%, otherwise the candidate is discarded and logged. The campaign uses the `grtresna_bicomplex_scalar` matter model, which carries two independent complex scalar fields—one canonical, one phantom—so that the per-lump gravitational signs in Eq. (2) are preserved in both the `GRTresna` constraint solve and the `GRTeclyn` evolution stress tensor $T_{\mu\nu}$. This eliminates the initial-data / evolution sign mismatch that afflicted earlier single-field campaigns. The post-load Hamiltonian and momentum gate, together with bounded CCZ4 constraint growth during evolution, therefore establishes both numerical control and source-model consistency for the bicomplex sector.

H. Scoring: how every evaluation is rewarded

Every evaluation in the campaign is reduced to one fitness value $S(\theta)$ by the same fixed rule, so candidates are ranked consistently. Each named quantity below is a normalized diagnostic in $[0,1]$ (penalties are negative); the score is their weighted sum,

$$\begin{aligned}S = &1000\, f_{\rm geo}^{\rm evol} + 1000\, f_{\rm geo}^{\rm op} + 600\, P_{\rm ftl} + 200\, F_{\rm op}\\ &+ 200\, C_{\rm curv} + 5\, G_{\rm nt} + g_{\rm h}\, \Sigma_{\rm health}\\ &+ 40\, w_x\, X_{\rm exotic} + 200\, H_{\rm hor},\end{aligned} \tag{6}$$

with the health block

$$\Sigma_{\rm health} = 150\,{\rm surv} + 60\,E + 40\,{\rm stab} + 30\,{\rm cmv} + 10\,\mathcal{H}_{\rm ok} + 15\,I, \tag{7}$$

TABLE I. Reward terms in the campaign objective [Eq. (6)], applied identically to every MAP-Elites evaluation.

symbol	rewards / penalizes	weight
$f_{\rm geo}^{\rm evol}$	4D evolving null shortcut	1000
$f_{\rm geo}^{\rm op}$	geodesic (frozen) shortcut	1000
$P_{\rm ftl}$	FTL persistence (gated)	600
$F_{\rm op}$	operational coord. FTL (gated)	200
$C_{\rm curv}$	curvature activity	200
$G_{\rm nt}$	nontrivial geometry	5
surv	structural survival	$150\, g_{\rm h}$
E	energy-condition health	$60\, g_{\rm h}$
stab	metric stability	$40\, g_{\rm h}$
cmv	co-moving stability	$30\, g_{\rm h}$
I	instability penalty	$15\, g_{\rm h}$
$\mathcal{H}_{\rm ok}$	constraint health	$10\, g_{\rm h}$
$X_{\rm exotic}$	exotic-matter penalty	$40\, w_x$
$H_{\rm hor}$	horizon-formation penalty	200

gated by the nontriviality factor $g_{\rm h} \in [0,1]$ so that a trivial (flat) configuration earns no health reward. Table I lists what each term measures. Two design choices matter. First, transport dominates: the evolving 4D shortcut $f_{\rm geo}^{\rm evol}$ and its geodesic counterpart carry the largest weights, so the score chiefly tracks genuine null-transport gain. Second, the persistence and operational terms $P_{\rm ftl}, F_{\rm op}$ are *dispersion-gated*—multiplied by the fraction of matter still confined—so a candidate cannot bank a coordinate superluminal channel after its lumps have flown apart. The exotic penalty $X_{\rm exotic} < 0$ charges for energy-condition violation at campaign weight w_x. All single scores quoted later are values of this S on a $t = 16$ evolution: search rankings, not physical observables, and never compared across runs of different duration.

For the CMA-ES champion (candidate 146) the search-tier score is $S = 852.15$, dominated by evolving-shortcut and gated-persistence terms. The CMA-ES refinement lifted the score from the MAP-Elites plateau ($S = 825.19$, candidate 87) primarily through improved confinement-gated fitness ($0.396 \to 0.417$), not a larger raw geodesic excess (the search-tier $f_{\rm geo}$ actually decreased from 25.3% to 20.1%). This illustrates the dispersion gate at work: CMA-ES found a genome that trades a smaller instantaneous shortcut for better structural persistence.

I. Search and validation stages

MAP-Elites and CMA-ES evaluate $\mathcal{O}(10^2)$ genomes, so the discovery loop uses a reduced GRTeclyn evolution configuration for wall-clock speed. Campaign env variables `GRTRESNA_EVOLUTION_L_FULL`, `GRTRESNA_EVOLUTION_N_FULL`, `GRTRESNA_EVOLUTION_MAX_LEVEL`, and `GRTRESNA_EVOLUTION_REGRID_THRESHOLD` set the search-

TABLE II. GRTeclyn evolution settings: search tier (MAP-Elites / CMA-ES) versus production validation (HQ / Richardson medium). Search uses a reduced grid for throughput; validation upgrades L, N, `max_level`, and $t_{\rm stop}$. Campaign env: `GRTRESNA_EVOLUTION_{L,N}_FULL`, `_MAX_LEVEL`, `_REGRID_THRESHOLD`.

parameter	search	HQ / RM
L	64	128
N	128	256
$h = L/N$	0.5	0.5
`max_level`	1	3
`regrid_threshold`	0.1	0.02
$t_{\rm stop}$	16	30

tier AMReX inputs `max_level` and `regrid_threshold` together with the evolution box (L, N) and $t_{\rm stop}$. Archive / CMA scores are therefore *search-tier* diagnostics, not continuum claims. Only the frozen champion is re-run at production resolution for validation.

- **Stage 1 — MAP-Elites discovery (fast search grid):** quality-diversity search at $L = 64$, $N = 128$ (grid spacing $h \equiv L/N = 0.5$), `max_level`$= 1$, `regrid_threshold`$= 0.1$, and $t_{\rm stop} = 16$ (bicomplex matter; per-lump gravitational signs retained in evolution). Coarse AMR is intentional: it buys enough evaluations to fill the behavior archive. Produces a diverse archive and a provisional score leader.

- **Stage 2 — CMA-ES refinement (same search grid):** warm-start from the MAP-Elites champion (and optionally its nearest elites); local covariance-matrix adaptation at the *same* $(L, N,$ `max_level`$, t_{\rm stop})$ and scoring rule. Only the refined top genome is promoted further.

- **Stage 3 — production replay (HQ):** the frozen CMA-ES champion is re-solved and evolved at higher fidelity / longer $t_{\rm stop}$—typically $L = 128$, $N = 256$, `max_level`$= 3$, `regrid_threshold`$= 0.02$, $t_{\rm stop} = 30$—with multiple emission times and multi-ray geodesic probes. Paper headline numbers come from this stage, not from MAP-Elites.

- **Stage 4 — resolution matrix:** matched coarse/medium/fine resolution runs (RC/RM/RF at fixed $L = 128$, $N = 192/256/384$) test whether the shortcut survives discretization changes at the production `max_level`. Domain-ladder and waveform runs are deferred.

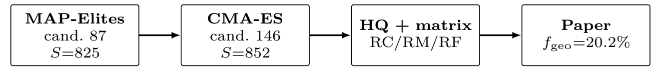

FIG. 3. Outer campaign ladder. MAP-Elites finds the diverse archive and champion (87); CMA-ES refines it to candidate 146; only the refined champion enters high-resolution replay and the resolution matrix.

TABLE III. Two-stage search hand-off: MAP-Elites champion (87) → CMA-ES refined champion (146) → HQ validation (RM). Search-tier grid: L=64, N=128, `max_level`=1, t=16. HQ grid: Table II.

stage	S	rays	peak $f_{\rm geo}$	gated fitness
MAP-Elites (87)	825.19	3/3	25.35%	0.396
CMA-ES (146)	852.15	3/3	20.12%	0.417
HQ RM (146)	—	5/5	20.21%	—

III. SEARCH METHODOLOGY

A. MAP-Elites quality-diversity

MAP-Elites [7] is a quality-diversity search: instead of returning one champion, it maintains an 8×8 archive with at most one elite per behavior cell. A candidate displaces the occupant of its cell only if its score $S(\theta)$ [Eq. (6)] is higher. The two archive axes are peak shortcut strength and FTL lifetime, so the campaign can keep short-lived strong channels and longer weaker ones side by side. The bicomplex campaign scheduled 200 evaluations: 148 completed GPU evolution, 50 failed the post-load gate, one failed the GRTresna admission gate, and one failed the constraint solve. Nine cells were occupied (14.06% coverage). All nontrivial elites lie in the highest lifetime row, while peak strength spans the archive. The best score rose from 690.1 to 825.19 and last improved at batch 10, when candidate 87 entered cell $(7, 7)$.

B. Archive outcome and CMA-ES refinement

The bicomplex MAP-Elites winner (candidate 87, $S = 825.19$) combines a search-tier evolving shortcut of $f_{\rm geo} = 25.3\%$ with dispersion-gated fitness 0.396, constraint-growth score 0.983, and confinement retention 0.396. CMA-ES warm-started from this genome and ran 150 evaluations (144 GPU-ok) at the same search grid, raising the score to 852.15 (candidate 146, $\Delta S \approx +27$). The improvement is not a bigger raw shortcut—the search-tier $f_{\rm geo}$ actually *decreased* from 25.3% to 20.1%—but a better confinement-gated persistence (0.396 → 0.417). CMA-ES converged to a genome with nearly static lump placement: very slow angular velocities and small breathing, producing a quasi-static scaffold that keeps matter confined longer (Fig. 7).

Search-tier confinement already falls to ∼40% by $t =$

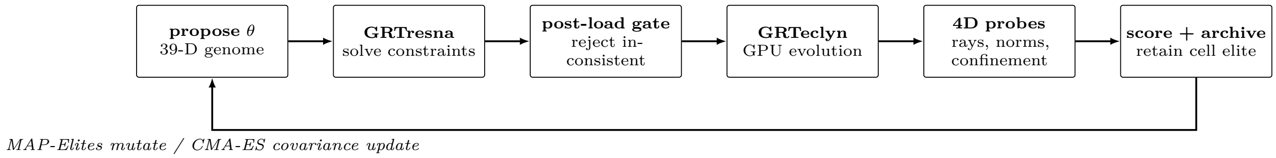

FIG. 4. Inner evaluation loop shared by MAP-Elites and CMA-ES. Failed constraint solves and post-load checks never enter GPU evolution; successful evaluations compete for archive cells (MAP-Elites) or update the CMA distribution.

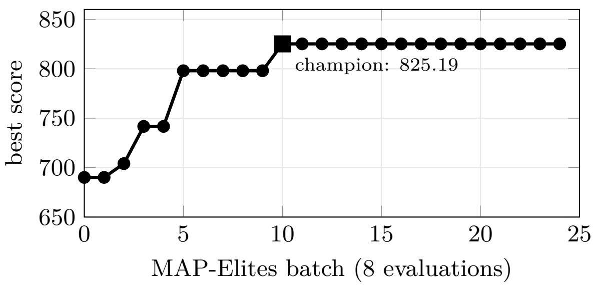

FIG. 5. MAP-Elites running best score over 200 evaluations. The champion (candidate 87) enters cell $(7, 7)$ at batch 10 with $S = 825.19$.

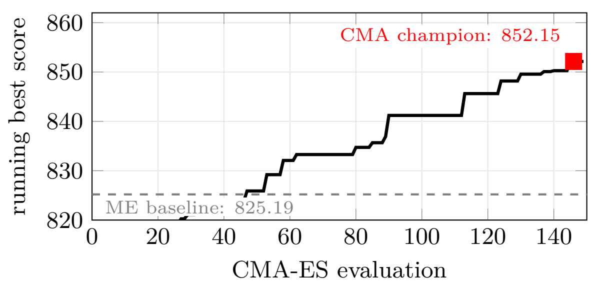

FIG. 6. CMA-ES refinement progress over 150 evaluations, warm-started from MAP-Elites champion 87. Dashed line: MAP-Elites best (825.19). The running best exceeds the baseline early and reaches 852.15 at candidate 146.

16, so longer production runs are expected to stress structural persistence even if the evolving null shortcut remains positive.

IV. DIAGNOSTICS AND PROBE VALIDATION

A. The 4D evolving null-geodesic trace

The transport claim rests on tracing light rays through the simulated spacetime, not on reading coordinate speeds off a slice. A coordinate operational shortcut $F_{\rm op}$ [Eq. (9)] can flag tilted light cones, but it is partly gauge dependent and can be inflated by a coordinate choice. The authoritative probe therefore integrates the null-geodesic equations

$$\dot{x}^\mu = g^{\mu\nu} k_\nu, \qquad \dot{k}_\mu = -\tfrac{1}{2}(\partial_\mu g^{\alpha\beta}) k_\alpha k_\beta, \tag{8}$$

for a photon with position x^μ and momentum k_μ. "4D" means the ray is not confined to one snapshot: it is pushed through a stack of stored time slices, so at each step the metric $g_{\mu\nu}(t, \mathbf{x})$ used is the one that actually exists when the photon is at that point. This distinguishes it from a *frozen* trace, which wrongly assumes a single slice persists for the whole flight.

The construction is as follows. A ray is emitted at time $t_{\rm emit}$ from a fixed source point toward a fixed receiver. The quantities that enter the shortcut formulas are *transit durations*, not absolute grid arrival times: $t_B^{\rm curved}$ (respectively $t_{\rm min}$) is the coordinate time elapsed along the curved path after emission, and $t_B^{\rm flat}$ (respectively $t_{\rm flat}$) is the matched flat-space flight duration between the same endpoints. Absolute arrival on the grid is $t_{\rm arrival} = t_{\rm emit} + t_B^{\rm curved}$. The fractional advantage is

$$F_{\rm op} = \frac{t_B^{\rm flat} - t_B^{\rm curved}}{t_B^{\rm flat}}, \tag{9}$$

$$f_{\rm geo} = \max\left(0,\ \frac{t_{\rm flat} - t_{\rm min}}{t_{\rm flat}}\right), \tag{10}$$

so $f_{\rm geo} = 0$ means no shortcut and $f_{\rm geo} = 0.20$ means the curved transit is 20% shorter than the flat comparison. A launch is accepted only if its rays stay physical: the relative drift in the null Hamiltonian $g^{\mu\nu} k_\mu k_\nu = 0$ must remain below 10^{-2}, which rejects rays that have accumulated interpolation error. The simulation itself runs to $t = 30$, but that is the evolution horizon of the metric stack, not the emission schedule. A ray launched at $t_{\rm emit}$ still needs a long remaining segment of evolved geometry to reach the far endpoint (flat transit is $t_{\rm flat} = 14.4$), so late launches near $t = 30$ cannot finish a fair comparison. We therefore sweep seven early launch times, $t_{\rm emit} = 0, 2, \ldots, 12$ (five rays each), and report the largest accepted $f_{\rm geo}$—the best moment to "surf" the channel. Across the validation matrix the peak sits at $t_{\rm emit} = 4$ (just after the PD pump switches off) and has already fallen by $t_{\rm emit} = 12$, so the window covers the opening of the channel rather than cutting it off.

B. Probe controls and calibration

Three controls bracket the probe's dynamic range before any candidate is evaluated.

- **Minkowski (flat):** $f_{\rm geo} = 0$ by construction; verifies that the tracer introduces no spurious shortcut on trivial data.
- **Prescribed Alcubierre** [4]**:** a moving-bubble metric ($v_s = 0.5$, bubble radius 2, wall width 2) is built analytically and fed to the evolving tracer, yielding $f_{\rm geo} \approx 32\%$ with 5/5 rays reaching. This confirms that the 4D probe recovers a known coordinate shortcut.
- **Negative control (eval 086):** an earlier campaign candidate whose frozen-slice diagnostic shows $f_{\rm geo} \approx 5.8\%$, but whose 4D evolving HQ trace collapses to 0% (0/5 rays). This false positive motivated the design rule that the evolving trace is authoritative: frozen peaks alone do not certify transport.

The positive campaign control is the CMA-ES champion (candidate 146): every accepted launch in all three resolution-matrix runs has 5/5 rays reaching and passes the null-Hamiltonian gate.

V. RESULTS: CANDIDATE-146 WARP PRECURSOR

A. Provenance and genome

Candidate 146 is the CMA-ES-refined champion promoted from the bicomplex MAP-Elites campaign. MAP-Elites saturated at candidate 87 ($S = 825.19$); CMA-ES warm-started from that genome and after 150 evaluations raised the score to $S = 852.15$ (candidate 146, $\Delta S \approx +27$). Its provenance freezes the exact 39-dimensional genome, source checksums, solver parameters, and source commit. The configuration uses the `grtresna_bicomplex_scalar` matter model with per-lump signs preserved through both the constraint solve and the GPU evolution. The resolution-matrix RC and RM runs were metrics-only; the RF (fine) run was evolved with `GRTECLYN_FRAMES=1` and fixed color scales, from which field-map movies were produced.

B. Matter dynamics

Candidate 146 uses five lumps with source signs $s_k = (+1, -1, -1, +1, -1)$: two canonical and three phantom (exotic). The collective breathing is small ($A_{\rm b} = 0.083$, $\omega_{\rm b} = 0.429$), the vertical oscillation is moderate ($A_z = 2.32$, $\omega_z = 0.016$), and the initial radii are $R_0 = (3.69, 5.50, 2.83, 5.59, 2.61)$. All angular velocities are negative, $\omega_{\rm rot} = (-0.0017, -0.0007, -0.0031, -0.0009, -0.0049)$, so all five trajectories are retrograde but extremely slow. Radial drifts are small [$v_r = (-0.060, -0.010, 0.001, -0.005, -0.006)$]; the slow rotation and drift—roughly an order of magnitude quieter than MAP-Elites champion 87—is what CMA-ES converged to in order to maximize confinement-gated persistence.

Figure 7 shows these prescribed support-center trajectories [Eq. (4)] at four times. The Q-ball fields are initially boosted along the corresponding motion. The PD trajectory controller (Sec. II F) drives the lumps during the ignition phase ($t < 4$) and then switches off. The bounded constraints ($\|\mathcal{H}\|_2 \leq 4.73 \times 10^{-3}$, $\|\mathcal{M}\|_2 \leq 9.16 \times 10^{-4}$ in RM) show that the evolution remains numerically controlled. The support centers are the intended dynamical scaffold, not tracked density isosurfaces. The field follows them imperfectly: confinement falls from 72% to 11% and rms spread grows by a factor 3.27 by $t = 30$. Candidate 146 is therefore a transient, initially driven configuration, not a bound five-body soliton.

C. Transport mechanism

The shortcut arises from the interplay of matter momentum and geometry. Each Q-ball lump oscillates with $\Pi \neq 0$ [Eq. (1)], sourcing a momentum density S_i that enters the momentum constraint and generates a nontrivial shift β^i. The shift tilts the local light cones in the direction of the matter current; null rays propagating through this region accumulate a coordinate-time advance relative to the flat reference path (Fig. 10). All five orbits in candidate 146 are retrograde ($\omega_{\rm rot} < 0$) but extremely slow ($|\omega_{\rm rot}| \lesssim 5 \times 10^{-3}$), so the scaffold is quasi-static: CMA-ES traded a larger instantaneous shortcut (the MAP-Elites champion had $f_{\rm geo} = 25.3\%$ at search tier) for better confinement-gated persistence ($0.396 \to 0.417$), selecting slow orbits that keep matter confined longer.

The peak $f_{\rm geo}$ occurs at $t_{\rm emit} = 4$, precisely when the PD pump switches off and the surviving confined scaffold begins its free evolution (Sec. II F). At later emission times, the matter has dispersed further and the channel weakens—by $t_{\rm emit} = 12$ the shortcut is substantially reduced (Fig. 8). Configurations that scored lower in the archive failed for the same reason: their supporting matter dispersed and the transport channel closed with it (Sec. VI A). The dispersion gate in the scoring function [Eq. (6)] explicitly penalizes this failure mode, steering the search toward genomes that maintain confinement.

D. 4D falsification certificate

The headline claim is operational: in the RM run, null rays between the fixed endpoints arrive earlier than in flat space. The evolving-geodesic artifact reports:

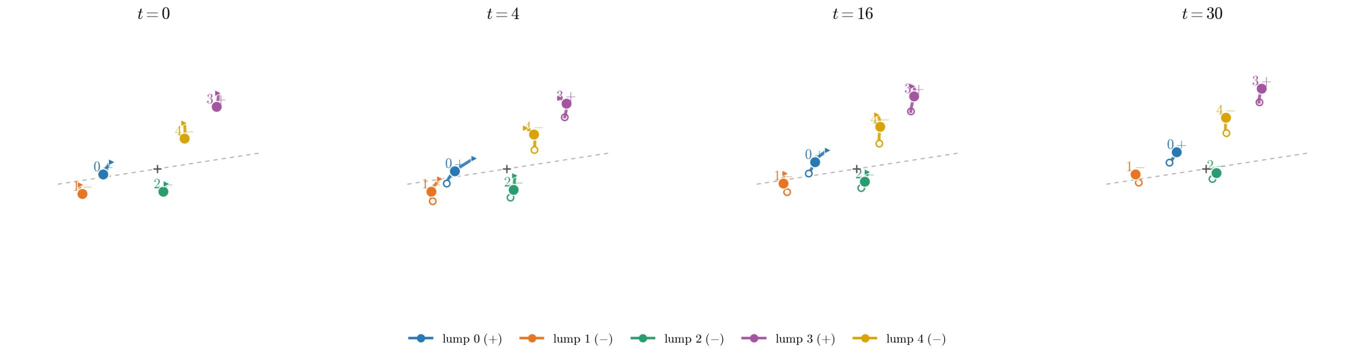

FIG. 7. Three-dimensional dynamics encoded by the frozen candidate-146 genome. Panels show the five prescribed PD-controller centers at $t = 0, 4, 16$, and 30. Filled circles are the centers at the stated times. To make the small motion legible, each local trajectory glyph covers four code-time units before and after the snapshot (where available), with its displacement about the filled center magnified by $4\times$; the hollow circle and arrow mark the start and end of that window. Colors and labels identify the lumps (+ canonical, − phantom), and the dashed line marks the x-directed null-probe axis. The magnification is representational: the centers are plotted at their unscaled positions. CMA-ES converged to very slow angular velocities ($|\omega_{\rm rot}| \lesssim 5 \times 10^{-3}$) and small breathing ($A_{\rm b} = 0.083$), producing a quasi-static scaffold that maximizes confinement-gated persistence.

peak 4D $f_{\rm geo}$	20.21%
frozen-peak $f_{\rm geo}$	29.45%
rays reached	5/5
$t_{\rm emit}$	4.00
$t_{\rm arrival}$	15.49
$t_{\rm min} = t_{\rm arrival} - t_{\rm emit}$	11.49
$t^{\rm flat}_{\rm transit}$	14.40
$h_{\rm quality\_ok}$	true ($h^{\rm max}_{\rm rel} = 2.19 \times 10^{-3}$)
search-tier peak (CMA-ES)	20.12% (3/3 rays)

Explicitly, $(t_{\rm flat} - t_{\rm min})/t_{\rm flat} = (14.40 - 11.49)/14.40 = 20.21\%$, matching Eq. (10). The peak emission time $t_{\rm emit} = 4$ coincides with the moment the PD pump switches off, suggesting that the transport channel opens as the driven ignition ceases and the matter begins its free evolution. The production HQ value agrees with the search-tier CMA-ES value to within 0.5% relative, confirming that the coarse search grid captures the essential physics. The endpoint timing corresponds to an effective $c/(1 - f_{\rm geo}) \simeq 1.25c$ relative to the matched flat path; locally the ray remains null.

E. Resolution matrix

The completed validation matrix is a three-run resolution ladder at fixed domain $L = 128$: coarse RC ($N = 192$, $h = 0.667$), medium RM ($N = 256$, $h = 0.500$), and fine RF ($N = 384$, $h = 0.333$). RM is the headline HQ baseline; paper numbers come from this run. No domain ladder (DS/DL) was run for candidate 146; that check is deferred to future work.

Across the resolution ladder, the same genome shortens null travel by 20.19–20.44% relative to flat space. The fine–medium relative difference is $(20.21 - 20.19)/20.21 =$

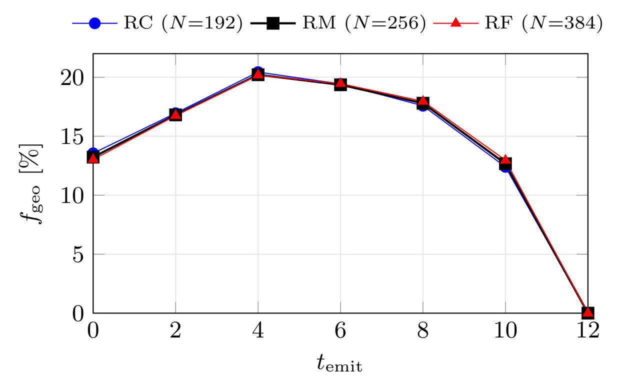

FIG. 8. Evolving null-geodesic emission sweeps ($t_{\rm emit} = 0, 2, \ldots, 12$) for all three resolution-matrix runs; the metric evolves to $t = 30$, but late launches lack enough remaining geometry for a complete transit. All runs peak at $t_{\rm emit} = 4$, just after PD pump-off, and agree within 0.26 percentage points. Every point shown has 5/5 rays reaching and passes the null-Hamiltonian quality gate.

0.1%, far below the preregistered 10% matrix tolerance. The frozen-peak $f_{\rm geo}$ is similarly stable at $\sim$29.5% across all three resolutions. For the $L = 128$ domain, the $t = 30$ stop precedes the arrival of boundary-reflected gauge modes at the central region: in 1+ log slicing those modes propagate at characteristic speeds up to $\sim\sqrt{2}\,c$, so a one-way crossing from the outer boundary requires $t \gtrsim L/(2\sqrt{2}) \approx 45$.

For unequal spacings, a Richardson estimate would assume

$$Q(h) = Q_0 + Ch^p, \qquad \frac{Q_c - Q_m}{Q_m - Q_f} = \frac{h_c^p - h_m^p}{h_m^p - h_f^p}. \tag{11}$$

The sequence $20.44\% \to 20.21\% \to 20.19\%$ is monotonically decreasing, consistent with convergence. The fine–

TABLE IV. Candidate-146 resolution matrix. All runs use $L = 128$, three AMR levels (`max_level`= 3, `regrid_threshold`= 0.02), and stop at $t = 30$. Peak $f_{\rm geo}$ is the best evolving 4D shortcut; frozen peak is the maximum of the frozen-slice $f_{\rm geo}$ time series.

run	N	h	peak $f_{\rm geo}$	frozen peak	$t_{\rm emit}$
RC	192	0.667	20.44%	29.15%	4
RM	256	0.500	20.21%	29.45%	4
RF	384	0.333	20.19%	29.54%	4

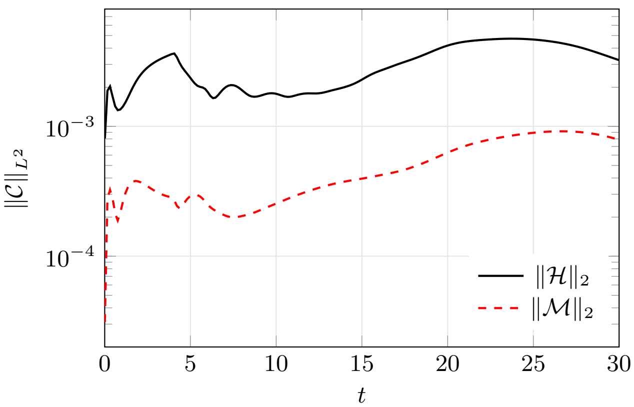

FIG. 9. Constraint norms in the headline RM run. The complete 3000-step diagnostic is plotted from a decimated table for typesetting only; maxima are evaluated from the full data.

medium difference is so small (0.02% absolute) that the extracted convergence order would be dominated by numerical noise. We therefore accept continuum stability of the 20.2% result without quoting a formal extrapolated value or grid-convergence index.

F. Evolution constraints

For RM, the volume-weighted norms remain below $\|\mathcal{H}\|_2 = 4.73 \times 10^{-3}$ and $\|\mathcal{M}\|_2 = 9.16 \times 10^{-4}$ through $t = 30$ (Fig. 9). These are comparable to the search-tier values and remain bounded throughout the evolution. The constraint norms peak during the early driven phase ($t \lesssim 4$) and then plateau or slowly grow during the free evolution, consistent with the dispersing matter producing less active refinement at late times.

G. Frozen versus evolving shortcut, and confinement

RM's frozen-peak $f_{\rm geo}$ reaches 29.45%, compared with the evolving 4D sweep peak of 20.21%. The ∼9% gap is physical: the frozen trace assumes that one snapshot persists for the entire ray flight, whereas the 4D trace samples the actual changing metric and therefore "sees" the channel open and then weaken during propagation. Only the latter is used for the transport claim. The frozen peak is stable across all three resolutions (29.1–29.5%), confirming that the metric snapshot quality is not resolution-sensitive.

The matter structure is not durable. In RM, confined matter fraction falls from 72% at $t = 0$ to 11% by $t = 30$, while the rms spread grows by a factor 3.27. The peak shortcut at $t_{\rm emit} = 4$ occurs just as the pump switches off, and the evolving $f_{\rm geo}$ curve falls steadily at later emission times. Successful ray transport therefore coexists with substantial dispersal: the channel is transient.

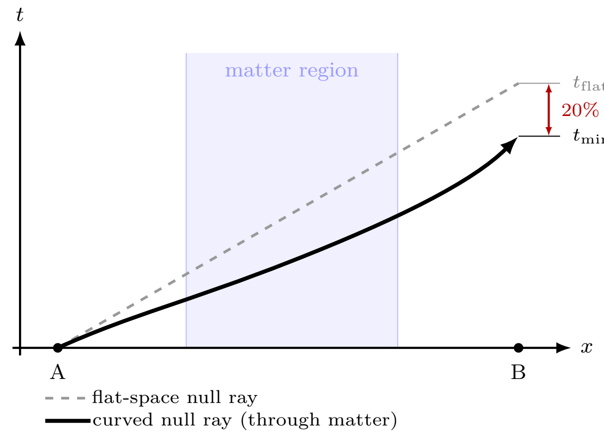

FIG. 10. Spacetime diagram of the null-geodesic shortcut. A photon emitted at A propagates toward receiver B. In flat space (dashed) the transit takes $t_{\rm flat}$. Inside the Q-ball matter region (shaded band) the geometry bends null paths so that the ray through the curved spacetime (solid) arrives at B in $t_{\rm min} < t_{\rm flat}$. The bracket on the right marks the ∼20% coordinate-time saving. The ray never leaves its local light cone.

H. What moves faster than light

Nothing physical moves outside its local light cone. The quoted ∼20% is a *null-geodesic timing advantage* through the evolved geometry relative to a flat reference path between the same endpoints. Coordinate light-speed maps visualize the tilted cones but do not establish the result by themselves. Figure 10 illustrates the mechanism schematically.

VI. DISCUSSION

A. The dispersion problem

Candidate 146 should be read against the many configurations that did not work. Across roughly a dozen ansatz families and several full MAP-Elites campaigns we repeatedly found the same failure mode: the supporting matter disperses and the transport channel closes with it. Spherical-harmonic shells yielded a single weak hit in ∼200 evaluations. Free massive scalar packets have no bound state and blow apart almost immediately. Bicomplex phantom lumps opened a gauge-invariant shortcut but lost ∼98% of their structure by $t = 16$ (structural persistence ≈ 0.02). Boosted and stiffened Q-ball variants improved early confinement but still sheared apart once placed on an orbit. This is why the score explicitly gates transport by confinement (Sec. II H): without it, the optimizer is rewarded for a coordinate superluminal channel produced by matter that has already flown apart. CMA-ES improved confinement-gated persistence relative to the MAP-Elites champion ($0.396 \to 0.417$), but candidate 146 still retains only about 11% of its confined matter by $t = 30$ at HQ. The search-tier confinement (∼40% at $t = 16$) falls more steeply under the longer, higher-resolution production evolution. The honest summary is that a *transient* shortcut is reproducible and robust, while a *long-lived*, well-confined channel remains an open problem.

B. Applications and next steps

The pipeline supports (i) inverse design of null transport under energy-condition penalties, (ii) QD atlases of mechanism families, and (iii) Pareto maps of shortcut versus exotic cost and persistence. Immediate priorities are to run a domain ladder (DS/DL) to verify boundary insensitivity, add a pump-work diagnostic to quantify the energy deposited during the ignition phase, and extend the evolution to longer $t_{\rm stop}$ to track late-time dispersal. Longer-term goals include optimizing for non-dispersing matter models and confinement world-tube objectives.

C. Computational cost

All runs were performed on a single 8-GPU node (NVIDIA H100 80 GB HBM3). MAP-Elites and CMA-ES each dispatch one `GRTeclyn` evolution per GPU in parallel, with concurrent `GRTresna` constraint solves running on CPU (MPI, 8 ranks per solve). A 200-evaluation MAP-Elites campaign at search resolution ($N = 128$, $t_{\rm stop} = 16$) completes in approximately two wall-clock hours; the 150-evaluation CMA-ES refinement takes a similar time. The production validation stage is sequential: the headline RM run ($N = 256$, $t_{\rm stop} = 30$, three AMR levels) fits on a single GPU; the fine Richardson run RF ($N = 384$) required three GPUs due to memory constraints. Total wall-clock time from the start of MAP-Elites to the completion of the three-run resolution matrix was approximately ten hours.

VII. CONCLUSION

We presented an automatic matter-first discovery pipeline: MAP-Elites proposes scalar configurations, CMA-ES locally refines the archive champion, the constraints are solved, the spacetime is evolved, and HQ 4D null geodesics ask whether light arrives earlier than in flat space. MAP-Elites saturated at candidate 87 ($S = 825.19$); CMA-ES improved the search score to 852.15 (candidate 146), a lift driven by confinement-gated persistence rather than a larger raw shortcut. The CMA-ES champion, a bicomplex five-lump Q-ball shell, yields $f_{\rm geo} = 20.2\%$ in the headline RM run—light 20% earlier—and 20.2–20.4% across the three-run resolution ladder (fine–medium relative difference $\ll 1\%$). The frozen-peak $f_{\rm geo}$ reaches ∼29.5%, stable across resolutions. All rays pass quality gates (5/5 reaching; matrix-wide $h_{\rm rel}^{\rm max} \le 3.8 \times 10^{-3}$, with 2.19×10^{-3} in the headline RM run). The bicomplex matter model preserves per-lump canonical/phantom signs through both the constraint solve and the evolution. The PD controller switches off at $t = 4$; because the headline geodesic launches at $t_{\rm emit} = 4$, the 20.2% advantage is measured in freely evolving Einstein–Klein–Gordon dynamics. Confinement falls to ∼11% by $t = 30$, so the shortcut is transient. Candidate 146 is therefore a reproducible, continuum-stable numerical warp precursor produced by an automated two-stage search, not a continuum-extrapolated or functional warp drive.

VIII. CODE AND DATA AVAILABILITY

The project repository contains the campaign configuration and archive, candidate-146 frozen provenance (under `runs/grtresna_promote/sources/`), the three completed resolution-matrix validation summaries (RC/RM/RF evolving geodesic artifacts, constraint histories, and confinement timeseries). Figure files live in `research/neuralspacetime/figures/`; the article directory holds the reduced numerical tables used by TikZ/PGFPlots. MAP-Elites and CMA-ES trajectory files are under `runs/grtresna_qd/` and `runs/grtresna_cmaes/`.

IX. ACKNOWLEDGEMENTS

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